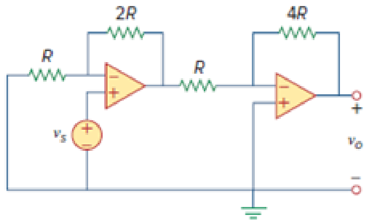
Chapter 5, Problem 59P

Problem

In the op amp circuit of Fig. 5.86, determine the voltage gain v_o/v_s. Take R = 10 kΩ.

Figure. 5.86:



Step-by-step solution

Step 1 of 2

Draw the modified circuit as shown in Figure 1.

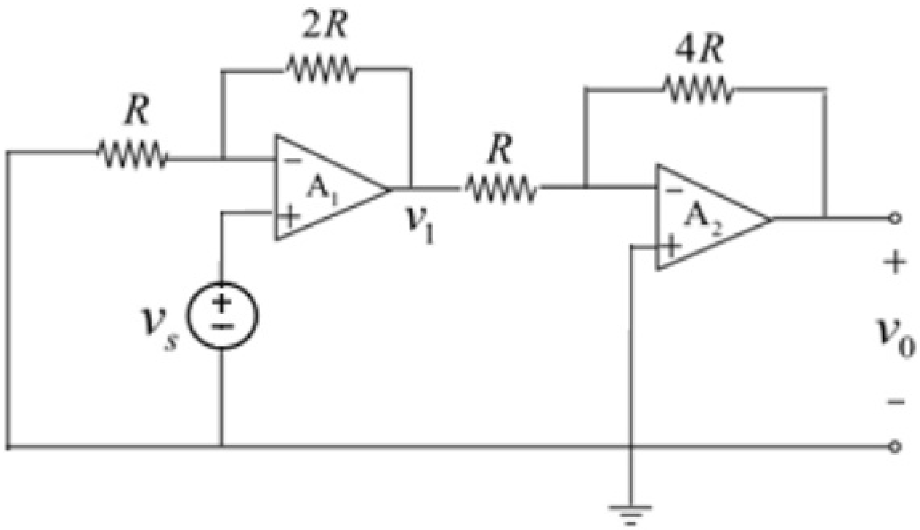


Figure 1

Step 2 of 2

The first op amp circuit is a non-inverting amplifier.

The expression of the output voltage of the first op amp is,

$$v_1 = \left(1 + \frac{2R}{R}\right) v_s$$

$$v_1 = 3v_s$$

The second op amp is an inverting amplifier.

The expression of the output voltage of the second op amp is,

$$\begin{aligned} v_o &= -\frac{4R}{R} v_1 \\ &= -4v_1 \end{aligned}$$

Substitute $3v_s$ for v_1.

$$v_o = -4(3v_s)$$

$$\frac{v_o}{v_s} = -12$$

Thus, the voltage gain $\frac{v_o}{v_s}$ is $\boxed{-12}$.